

Advanced techniques in applied economics

Lecture 3: DAGs, interventions, and structural equation models

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Spring 2026

From undirected structure to causal structure

What changes today?

In Lecture 2, a graph summarized **conditional independence**. Today we move to **directed** graphs, where arrows suggest an asymmetric data-generating structure.

Undirected graphs

Useful for describing conditional dependence networks. They are symmetric and descriptive.

Directed acyclic graphs

Useful for expressing causal or other directional assumptions.

Main question: Which variables matter for which others, and how would the system react to intervention?

Roadmap for today

1. DAGs and factorization
2. d-separation: chain, fork, collider
3. moralization as a way to test d-separation
4. Markov equivalence and CPDAGs
5. interventions and mechanism changes
6. linear SEMs and Wright's path tracing rules

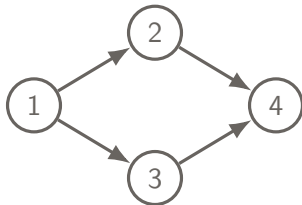
Memorable statement: a DAG is both a probabilistic model and a structural story.

Part 1: DAGs as structural models

What is a DAG?

A **directed acyclic graph** (DAG) consists of

- vertices $1, \dots, m$ representing variables $X_1, \dots, X_m$,
- directed edges $i \rightarrow j$,
- no directed cycles.



Interpretation

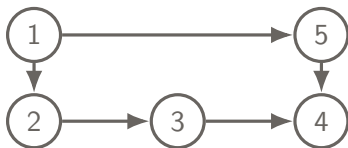
The arrow $i \rightarrow j$ says that X_i is a direct input into the mechanism generating X_j .

Parents, children, ancestors, descendants

Let D be a DAG.

For a node j :

- $\text{pa}(j)$ = parents of j ,
- $\text{ch}(j)$ = children of j ,
- $\text{an}(j)$ = ancestors of j ,
- $\text{de}(j)$ = descendants of j .



For node 4:

$$\text{pa}(4) = \{3, 5\}, \quad \text{an}(4) = \{1, 2, 3, 5\}.$$

For node 1:

$$\text{ch}(1) = \{2, 5\}, \quad \text{de}(1) = \{2, 3, 4, 5\}.$$

Factorization over a DAG

A distribution P is **Markov with respect to a DAG D** if its density factorizes as

$$p(x_1, \dots, x_m) = \prod_{j=1}^m p(x_j \mid x_{\text{pa}(j)}).$$

Interpretation

Each variable is generated from its parents and no other variables enter directly.

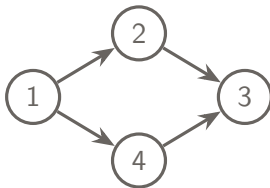
Why this matters

The graph now has a dual role:

- it determines a factorization of the joint law,
- it implies conditional independence statements.

A simple factorization example

Consider the DAG



Then the joint law factorizes as

$$p(x_1, x_2, x_3, x_4) = p(x_1) p(x_2 \mid x_1) p(x_4 \mid x_1) p(x_3 \mid x_2, x_4).$$

Think of X_1 as an initial state, X_2 and X_4 as intermediate channels, and X_3 as the final outcome.

In a DAG, each factor involves one variable and its parents.

From DAGs to structural equations

A DAG can be viewed not only as a factorization of a joint law, but also as a system of structural assignments.

Structural equation view

For each node j , write

$$X_j = f_j(X_{\text{pa}(j)}, \varepsilon_j), \quad j = 1, \dots, m,$$

where the noise variables $\varepsilon_1, \dots, \varepsilon_m$ are jointly independent.

Why this matches the DAG

Each variable depends only on its parents and on its own noise term. So the arrows describe which variables enter directly into each mechanism.

Local Markov property

For each node j in a DAG,

$$X_j \perp\!\!\!\perp X_{\text{nd}(j) \setminus \text{pa}(j)} \mid X_{\text{pa}(j)},$$

where $\text{nd}(j)$ denotes the non-descendants of j .

Interpretation

Once we know the parents of X_j , all non-descendants of j become irrelevant for predicting X_j .

Why this is natural

This says that all direct inputs into X_j have already been listed as parents.

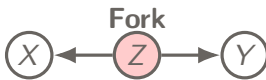
Part 2: d-separation

Three basic triplexes

Everything starts with three local patterns.



$$X \perp\!\!\!\perp Y \mid Z$$



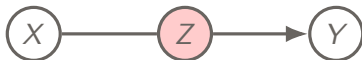
$$X \perp\!\!\!\perp Y \mid Z$$



$$X \perp\!\!\!\perp Y$$

Conditioning **blocks** a chain or fork, but **opens** a collider.

The chain pattern



Interpretation

Z is a transmission channel between X and Y . Once we condition on Z , the path is blocked:

$$X \perp\!\!\!\perp Y \mid Z.$$

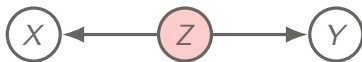
Policy example

Policy X changes investment Z , which changes output Y . After fixing investment, policy adds no further predictive content for output.

Finance example

Monetary news X moves interest rates Z , which then affect stock returns Y . After fixing interest rates, monetary news adds no further predictive content for returns.

The fork pattern



Interpretation

Z is a common cause of X and Y . Once we condition on Z , the path is blocked:

$$X \perp\!\!\!\perp Y \mid Z.$$

Demand-shock example

A common demand shock Z affects both price X and quantity Y . After controlling for the shock, the induced association disappears.

Ability example

Latent ability Z affects both education X and earnings Y . Conditioning on ability removes the spurious component of the association.

The collider pattern



Interpretation

Without conditioning, the path is blocked at the collider, so typically $X \perp\!\!\!\perp Y$.

Conditioning on Z opens the path and can create dependence (we discussed this in Week 1).

Selection example

Admission Z to a school or program may depend on both test scores X and extracurricular achievements Y . Among admitted students, these two variables may appear negatively related.

Labor-market example

Staying in a demanding occupation Z may depend on both ability X and motivation Y . Conditioning on those who remain in the occupation can induce an artificial dependence between them.

d-separation: the general rule

A path is **active given S** if

- every non-collider on the path is outside S ,
- every collider on the path has itself or one of its descendants in S .

Equivalently, a path is **blocked given S** if it is not active given S .

d-separation

Two sets A and B are **d-separated by S** if every path between A and B is blocked given S .

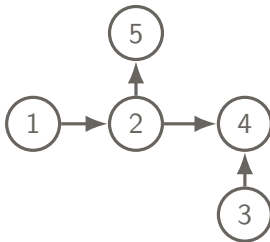
Markov property for DAGs

If A and B are d-separated by S in the DAG, then

$$X_A \perp\!\!\!\perp X_B \mid X_S.$$

A longer example

Consider the DAG



- 1 and 3 are d-separated by $\emptyset$ because the only path hits the collider 4.
- 1 and 3 become d-connected if we condition on 4.
- 1 and 4 are d-separated by $\{2\}$ because the chain is blocked at 2.

Moralization: a practical way to test d-separation

A useful way to test d-separation is to convert the DAG into an undirected graph.

1. Keep only the ancestors of $A \cup B \cup S$.
2. **Moralize**: connect any two parents of a common child.
3. Drop all arrow directions.
4. Check whether S separates A and B in the resulting undirected graph.

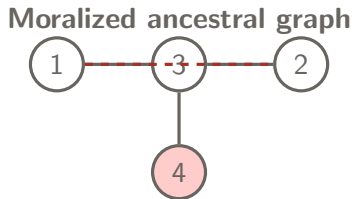
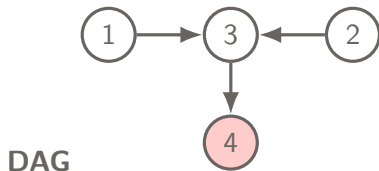
Moralization criterion

A and B are d-separated by S in the DAG if and only if S separates A and B in the moralized ancestral graph.

This reduces a d-separation question in a DAG to an ordinary separation question in an undirected graph.

Moralization on a collider example

We test whether 1 and 2 are d-separated given 4 in the DAG below:



Conclusion

Because 4 is a descendant of the collider 3, the path opens. Moralization connects the parents 1 and 2, so they are **not** d-separated given 4.

Part 3: Markov equivalence and CPDAGs

Markov equivalence: why observational data do not identify every arrow

Different DAGs can encode the same set of conditional independence relations.

Definition

Two DAGs are **Markov equivalent** if they imply the same conditional independence statements.



Equivalent example

These three DAGs are Markov equivalent. All of them imply $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

Non-equivalent collider

$\textcircled{1} \rightarrow \textcircled{2} \leftarrow \textcircled{3}$ is **not** equivalent to the three above, because it implies $X_1 \perp\!\!\!\perp X_3$.

From observational data, we can typically recover only a Markov equivalence class.

The Verma–Pearl characterization

Two DAGs are Markov equivalent if and only if they have

- the same **skeleton** (same undirected graph),
- the same **v-structures**.

v-structure

A triple of the form

$$X \rightarrow Z \leftarrow Y$$

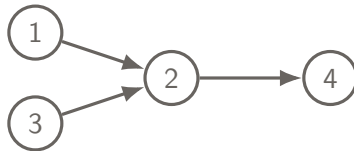
with no edge between X and Y .

Observational data identify the skeleton and the colliders, but often not all arrow directions.

CPDAGs / essential graphs

A **CPDAG** represents an entire Markov equivalence class.

- **Directed edges:** directions shared by every DAG in the class.
- **Undirected edges:** directions that cannot be identified from observational data alone.



Important point

All v-structures are fixed across the equivalence class, but there may be more arrows.

Example

Above $1 \rightarrow 2 \leftarrow 3$ is a v-structure but the arrow $2 \rightarrow 4$ is also determined.

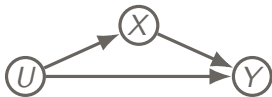
A CPDAG separates what is identifiable from observational data (with no extra assumptions) from what remains unresolved within the equivalence class.

Part 4: Interventions and policy changes

Observing is not the same as intervening

Observational data tell us how variables move together. Policy analysis asks a different question:

What happens if we change one of the mechanisms?



Example

X = policy variable

Y = economic outcome

U = background conditions

Observational question: What is $\mathbb{P}(Y \mid X = x)$?

Outcomes among units with observed value $X = x$.

Interventional question: What is $\mathbb{P}(Y \mid \text{do}(X = x))$?

This describes outcomes if we actively set X to x .

If U affects both X and Y , then

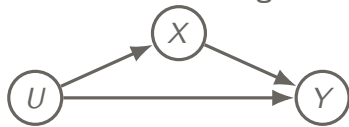
$$\mathbb{P}(Y \mid X = x) \neq \mathbb{P}(Y \mid \text{do}(X = x)).$$

Association describes the world as it is; intervention asks what happens if we change the system.

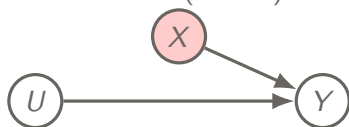
Hard interventions: the *do* operator

A hard intervention sets one variable externally: $\text{do}(X_k = x)$.

Observational regime



After $\text{do}(X = x)$



Graphical meaning

A hard intervention removes all arrows pointing into the intervened variable. So the mechanism generating X is replaced by an external assignment.

Economic reading

This is the idealized version of a randomized policy assignment: the policy variable is no longer chosen in response to background conditions.

Soft interventions: changing a mechanism

In economics, policy often changes a mechanism without fixing a variable to a constant.

Observational SEM

$$X = g(U, \varepsilon_X), \quad Y = h(X, U, \varepsilon_Y).$$

Soft intervention

A policy may change the equation for Y into

$$Y = h^*(X, U, \varepsilon_Y),$$

or change parameters inside h .

Examples

- a tax changes incentives,
- a subsidy shifts demand,
- monetary policy changes a reaction function,
- a regulation changes firms' decision rules.

Key idea

A soft intervention changes one structural mechanism while leaving most of the system intact.

Policy analysis is often about **mechanism shifts**, not just fixing variables by force.

How soft interventions can reveal causal direction

Suppose the true causal direction is $X \longrightarrow Y$, with structural equations

$$X = f(\varepsilon_X), \quad Y = g(X, \varepsilon_Y).$$

Now imagine a soft intervention that changes only the mechanism for X :

$$X = f^*(\varepsilon_X), \quad Y = g(X, \varepsilon_Y).$$

What changes?

Because the mechanism for X changes, generally

$$P^*(X) \neq P(X), \quad P^*(Y) \neq P(Y),$$

and typically also

$$P^*(X | Y) \neq P(X | Y).$$

What stays the same?

The mechanism from X to Y is unchanged, so

$$P^*(Y | X) = P(Y | X).$$

This is only an intuition, but it is the basic idea behind several methods based on invariance across environments.

A tiny simulation: invariant mechanism, changing input distribution

Suppose the true direction is

$$X \rightarrow Y, \quad Y = X^2 + \varepsilon_Y,$$

and only the distribution of X changes across environments.

```
set.seed(1)
n <- 1000

x0 <- rnorm(n, 0, 1)
y0 <- x0^2 + rnorm(n, sd = 0.5)

x1 <- rnorm(n, 2, 1)
y1 <- x1^2 + rnorm(n, sd = 0.5)
```

```
par(mfrow = c(1,2))
hist(x0, freq = FALSE, main = "X")
hist(x1, freq = FALSE, add = TRUE)

plot(x0, y0, pch = 16, col = 1)
points(x1, y1, pch = 16, col = 2)
```

Interpretation

The marginals change across environments, but the conditional mechanism relating Y to X remains the same.

Why histograms and conditional fits can reveal direction

In the simulation, the intervention changed the distribution of the cause X .

Across environments

- the histograms of X and of Y change,
- but the conditional relation $Y | X$ stays stable.

Wrong reverse direction

If we tried to model the system as $Y \rightarrow X$, we would need the mechanism $X | Y$ to remain stable across environments. In general this fails.

To identify direction, look for the orientation in which the conditional mechanism remains stable when environments change.

Part 5: Linear SEMs and path tracing

Linear structural equation models

A linear SEM associated with a DAG has the form

$$X_i = \sum_{j \in \text{pa}(i)} \beta_{ij} X_j + \varepsilon_i, \quad i = 1, \dots, m,$$

where $\varepsilon_1, \dots, \varepsilon_m$ are mutually independent (or at least satisfy $\mathbb{E}(\varepsilon_i \mid X_{\text{pa}(i)}) = 0$).

Matrix form: $X = BX + \varepsilon$, where $B_{ij} = 0$ when $j \notin \text{pa}(i)$.

Note

The zero pattern of B is exactly the DAG.

Interpretation of the example

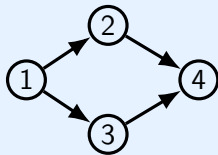
X_1 is a primitive shock, X_2 and X_3 are intermediate channels, and X_4 is the outcome of interest.

$$X_1 = \varepsilon_1,$$

$$X_2 = \beta_{21} X_1 + \varepsilon_2,$$

$$X_3 = \beta_{31} X_1 + \varepsilon_3,$$

$$X_4 = \beta_{42} X_2 + \beta_{43} X_3 + \varepsilon_4.$$



Gaussian LSEMs are exactly Gaussian DAG models

Recall: A DAG model is defined by the factorization $f(x) = \prod_{i=1}^m f(x_i \mid x_{\text{pa}(i)})$.

Gaussian DAG model

If X is jointly Gaussian, then each conditional distribution $X_i \mid X_{\text{pa}(i)}$ is Gaussian. So a Gaussian DAG model is specified by choosing these Gaussian conditionals.

Gaussian LSEM writes

$$X_i = \sum_{j \in \text{pa}(i)} \beta_{ij} X_j + \varepsilon_i, \quad \varepsilon_i \sim N(0, \omega_i),$$

where the errors ε_i are independent Gaussian.

Conditional law

$$X_i \mid X_{\text{pa}(i)} \sim N\left(\sum_{j \in \text{pa}(i)} \beta_{ij} X_j, \omega_i\right).$$

So a linear Gaussian SEM specifies exactly the Gaussian conditionals appearing in the DAG factorization.

Structural coefficients and total effects

Structural coefficients

In the linear SEM

$$X = BX + \varepsilon,$$

the matrix B records the **direct** linear effects encoded by the DAG.

Reduced-form map

Solving the system gives

$$X = (I - B)^{-1}\varepsilon.$$

So the matrix $(I - B)^{-1}$ combines **direct and indirect** effects.

Structural coefficients describe local mechanisms. The reduced-form map describes the total propagation of shocks.

Wright's path tracing: the basic idea

In a **linear** SEM, the covariance between two variables can be written as contributions from different paths in the graph.

Why this is useful: It helps explain an observed association as a combination of

- direct effects,
- indirect effects through mediators,
- common-cause effects.

Path tracing calculation

To understand $\text{cov}(X_1, X_4)$, we look at the two directed paths $1 \rightarrow 2 \rightarrow 4$, $1 \rightarrow 3 \rightarrow 4$. Each path contributes the product of its coefficients, so

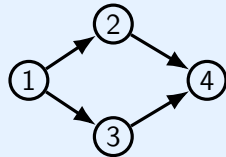
$$\text{cov}(X_1, X_4) = \beta_{21}\beta_{42} + \beta_{31}\beta_{43}.$$

$$X_2 = \beta_{21}X_1 + \varepsilon_2,$$

$$X_3 = \beta_{31}X_1 + \varepsilon_3,$$

$$X_4 = \beta_{42}X_2 + \beta_{43}X_3 + \varepsilon_4,$$

with $\text{var}(X_1) = 1$.



A caution about interpretation

DAGs and SEMs are powerful, but they do not identify themselves from thin air.

- d-separation gives testable implications,
- Markov equivalence limits what observational data can identify,
- structural interpretation always rests on substantive assumptions.

Important caution

A DAG is not automatically discovered just because we fit a model. It is a structured hypothesis about the data-generating process.

Graphs are useful because they make assumptions explicit, not because they eliminate the need for assumptions.